Divyaraj Chudasama

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.56
SHREE GANGOTRI VIDYAMANDIR HIGHSCHOOL	Class XII (Gujarat Secondary & High Secondary Education Board)	2020	90.46%
DR. P. V. DOSHI SARASWATI MADHAYAMIK VIDHAYALAY	Class X (Gujarat Secondary & High Secondary Education Board)	2018	89.00%

INDUSTRY INTERNSHIP EXPERIENCE

Effiya Technologies Pvt. Ltd. | Analyst Intern

- **Engineered ETL pipelines** to migrate **Abu Dhabi Commercial Bank's (ADCB)** surveillance platform to **SAS AML 8.3** across **Oracle and PostgreSQL**, implemented automated staging-to-cloud handshake validation jobs for daily transaction-level data loads.
- Configured **40+ transaction surveillance scenarios** (high-velocity funds, structuring, PEP) to flag financial crime patterns and automated **round-robin L1/L2 triage routing**.
- Developed **16 SAS Visual Analytics dashboards** for L1/L2 compliance teams, tracking active triage aging, customer journey timelines, and regulatory **Suspicious Transaction Reports (STRs)**.
- Implemented **SAS Viya ML alert scoring models** with automated **threshold hibernation** (<630 individual, <415 corporate), **filtering low-risk false positives** over a 6-month window while resurfacing critical alerts for repeat high-risk entities.

Qtech Analytics Pvt. Ltd. | AI/ML Intern

- **Developed an enterprise conversational RAG prototype** using **LangChain, FAISS, and Streamlit** to query multi-page financial PDF reports, implemented text parsing, recursive chunking, and **dense vector similarity** search to automate report analysis.
- **Tuned chunk sizes and top-k retrieval depth** to maximize context relevance, engineered strict **prompt-grounding constraints** and **conversational memory buffers**, reducing ungrounded responses and **halving manual information lookup time.**

PROJECTS

Autonomous Relational Analytics Agent | *Python, Pandas, Streamlit, Groq API, AST Sandbox, BFS, Plotly GitHub*

- **Architected an autonomous multi-agent analytics engine** with specialized **profiling, planning, coding, and storytelling agents** in a Streamlit UI with Plotly charts and execution traces, implemented **dual-model failover** (gpt-oss-120b to qwen3.8-27b on 429s) and an offline **heuristic compiler for zero-downtime resilience**.
- **Engineered autonomous schema discovery and multi-hop join pathfinding** using **Breadth-First Search (BFS)** across foreign keys to bridge non-adjacent tables in **complex star/snowflake schemas**, dynamically synthesized arbitrary **multi-column arithmetic expressions, filter predicates, and temporal period buckets with zero join hints.**
- **Built an AST execution sandbox with a 21-token injection blocklist** (os, subprocess, drop table), safe built-ins, 1,000-row cap, and an automated 3-iteration **self-healing loop,** achieved **100% pass rate (22/22 tests) on 15k rows,** eliminating naive ChatGPT's **silent row drops** and filter **hallucinations (which caused a 7.0% error).**

Credit Card Fraud Detection | *Python, scikit-learn, XGBoost, PyTorch, PyTorch Geometric GitHub*

- **Benchmarked 72 fraud-detection pipelines** across classical ML, boosting, **LSTM/Transformer, GAN oversampling and GNNs** on **284,807 transactions (0.17% fraud)** under a **chronological split** with in-pipeline resampling and validation-only tuning.

- **Delivered a Random Forest + SMOTE-ENN champion** with **PR-AUC 0.795** and **F1 0.848**, reaching **97.5% precision at 75% recall** with **1 false positive in 42,722** future transactions, served in a **Streamlit** app with per-prediction **SHAP** explanations.
- **Ran a 1,000× stratified bootstrap** (CIs 0.20-0.24, **all 15 pairs overlap**, ±0.020 PR-AUC/fraud band, not ranking), showed shuffling leaks **517 duplicates** (0 chronologically) and **pre-registered a published-benchmark replication** within **0.37 SE** (0.8055 vs 0.7902).

Distributed Rate Limiter | *Python, Redis, Lua, FastAPI, asyncio, pytest* *GitHub*

- **Engineered distributed rate-limiting middleware** with **token-bucket and sliding-window log/counter** as **atomic Redis Lua scripts** on Redis time, holding **exactly 20/20** per client over 3 load-balanced servers (5/5 runs) vs. **60 (3×)** in-memory, **33-48** racy GET/SET.
- **Hardened for failure modes** with **single-key, TTL-bounded scripts** (Redis Cluster-compatible), bounded pools, **0.5 s timeouts** and a fail-open/closed policy with an observable **bypass header**, after diagnosing fail-open silently admitting **150/150** on timeouts.
- **Verified with 57 pytest cases** on an injected clock (retry-after exact to **10 ms** at window edges; **300 concurrent requests from 5 clients exactly 25** allowed) and a **multi-process load harness** scaling to **5 instances × 500 requests**, holding **exactly 50/50**.

Multimodal RAG Platform | *Python, Django Ninja, Next.js, Qdrant HNSW, CLIP, BLIP, Qwen 27B, Celery, Docker* *GitHub*

- **Architected an enterprise multimodal RAG engine** with Django Ninja, Celery/Redis, and polyglot storage (MongoDB + Embedded Qdrant), built a Tailwind split-screen UI with **real-time SSE streaming** , deep **#page=N** citations ,and a graceful **degradation ladder**.
- **Engineered content-aware ingestion** using **pdfplumber** 2D extraction and **a PyMuPDF vector census** to capture **100% of schematics** (vs. 0% in naive extractors), enforced an **Atomic Whole-Table rule** that cut numeric hallucination by **31.2% (96.4% faithfulness)**.
- **Designed a 3-way hybrid retrieval & vision pipeline** fusing **MiniLM, TF-IDF, and CLIP ViT-B/32** via RRF (k=60), paired **BLIP captions** with **Qwen 27B** under **700×700** token budgeting, achieving **100% Hit@1, 1.000 MRR, and +16.4% answer relevance** (<20ms latency).

SKILLS

Programming Languages: Python, C/C++, SQL, Lua, Bash

Technologies & Systems: Docker, Redis, MongoDB, Qdrant (HNSW), PostgreSQL, Celery, Git, Linux

Frameworks & Libraries: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, FastAPI, Django Ninja, Pandas, NumPy, Streamlit, Plotly

AI/ML & Architecture: Multimodal RAG, Multi-Agent Systems, Vector Embeddings (CLIP, MiniLM), Hybrid Retrieval (RRF), GNNs, AST Sandboxing, Distributed Systems

SCHOLASTIC ACHIEVEMENTS

Secured **2nd** Place at Motive Hackathon at IIT-H

Secured **3rd** Place in the IIT-H Deep Learning 2026 Hackathon on Kaggle

Solved **500+** Data Structures and Algorithms problems across LeetCode and GeeksforGeeks

RELEVANT COURSES

Systems: Advanced Computer Networks, Network Security, Operating System , OOP, Database Management System

AI/ML: Foundations of Machine Learning, Deep Learning, Visual Computing , Fraud Analytics

Theory: Advanced Data Structures, Linear Optimization